

0a_EDA

April 10, 2026

1 1. Explorative Datenanalyse (EDA)

KI1-Projekt 308 — California Housing Datensatz

Ziel: Den Datensatz verstehen, Verteilungen und Korrelationen analysieren, Datenbereinigung dokumentieren.

```
[1]: import sys
sys.path.insert(0, '..')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from utils.data import load_raw_data, clean_data, load_and_clean_data
from utils.plotting import (
    plot_correlation_heatmap,
    plot_histograms,
    plot_features_vs_target,
    save_fig,
)

plt.rcParams['figure.dpi'] = 100
%matplotlib inline
```

1.1 1.1 Rohdaten laden und inspizieren

```
[2]: df_raw = load_raw_data()
print(f"Datensatz: {df_raw.shape[0]} Zeilen, {df_raw.shape[1]} Spalten")
print(f"\nFeatures: {list(df_raw.columns[:-1])}")
print(f"Zielvariable: {df_raw.columns[-1]}")
df_raw.head()
```

Datensatz: 20640 Zeilen, 9 Spalten

Features: ['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population',
'AveOccup', 'Latitude', 'Longitude']
Zielvariable: MedHouseVal

```
[2]:
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	\
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	

	Longitude	MedHouseVal
0	-122.23	4.526
1	-122.22	3.585
2	-122.24	3.521
3	-122.25	3.413
4	-122.25	3.422

```
[3]: df_raw.describe()
```

```
[3]:
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	\
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	
std	1.899822	12.585558	2.474173	0.473911	1132.462122	
min	0.499900	1.000000	0.846154	0.333333	3.000000	
25%	2.563400	18.000000	4.440716	1.006079	787.000000	
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	
max	15.000100	52.000000	141.909091	34.066667	35682.000000	

	AveOccup	Latitude	Longitude	MedHouseVal
count	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.070655	35.631861	-119.569704	2.068558
std	10.386050	2.135952	2.003532	1.153956
min	0.692308	32.540000	-124.350000	0.149990
25%	2.429741	33.930000	-121.800000	1.196000
50%	2.818116	34.260000	-118.490000	1.797000
75%	3.282261	37.710000	-118.010000	2.647250
max	1243.333333	41.950000	-114.310000	5.000010

```
[4]: df_raw.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   MedInc          20640 non-null  float64
1   HouseAge        20640 non-null  float64
2   AveRooms        20640 non-null  float64
3   AveBedrms       20640 non-null  float64
4   Population      20640 non-null  float64
```

```

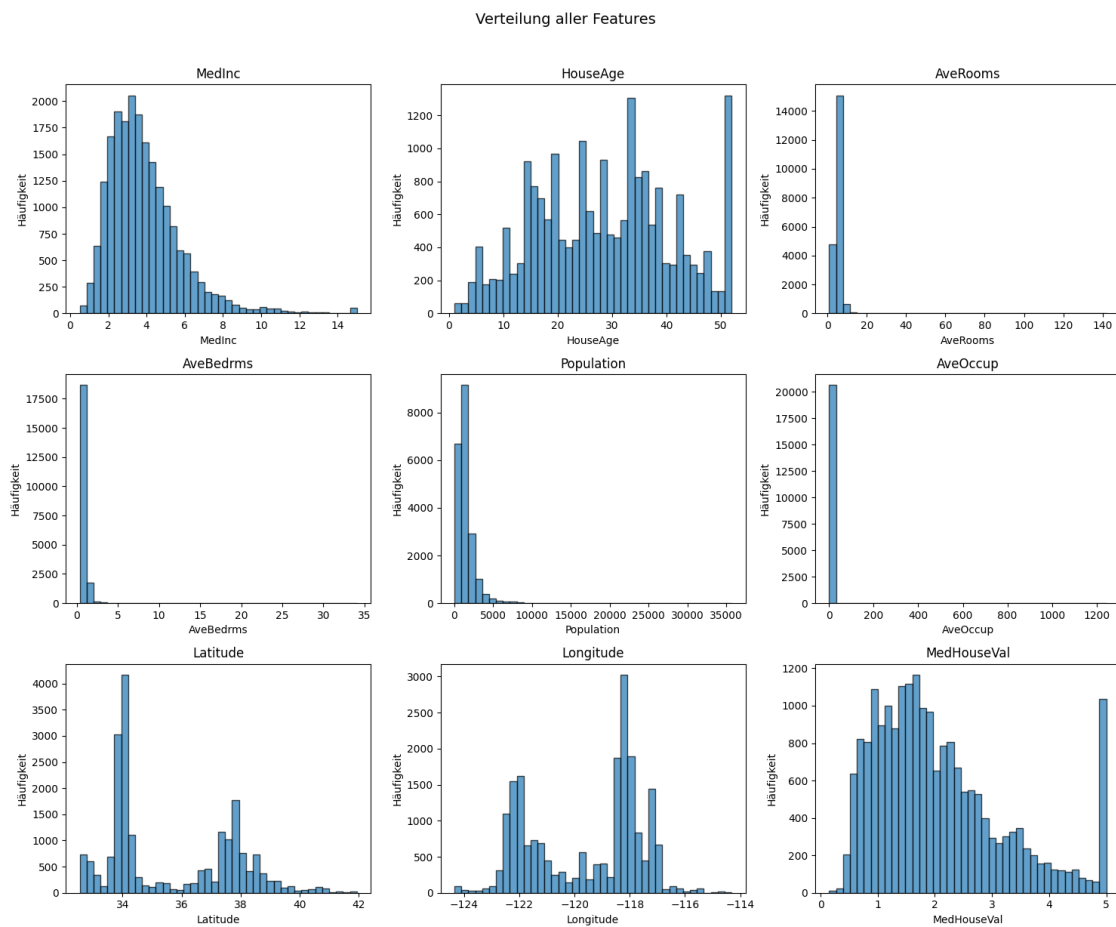
5 AveOccup      20640 non-null float64
6 Latitude      20640 non-null float64
7 Longitude     20640 non-null float64
8 MedHouseVal   20640 non-null float64
dtypes: float64(9)
memory usage: 1.4 MB

```

1.2 1.2 Verteilungen der Features (Histogramme)

```
[5]: fig, axes = plot_histograms(df_raw, bins=40, save_name="eda_histograms_raw")
plt.show()
```

Gespeichert: results/eda_histograms_raw.png und results/eda_histograms_raw.pdf

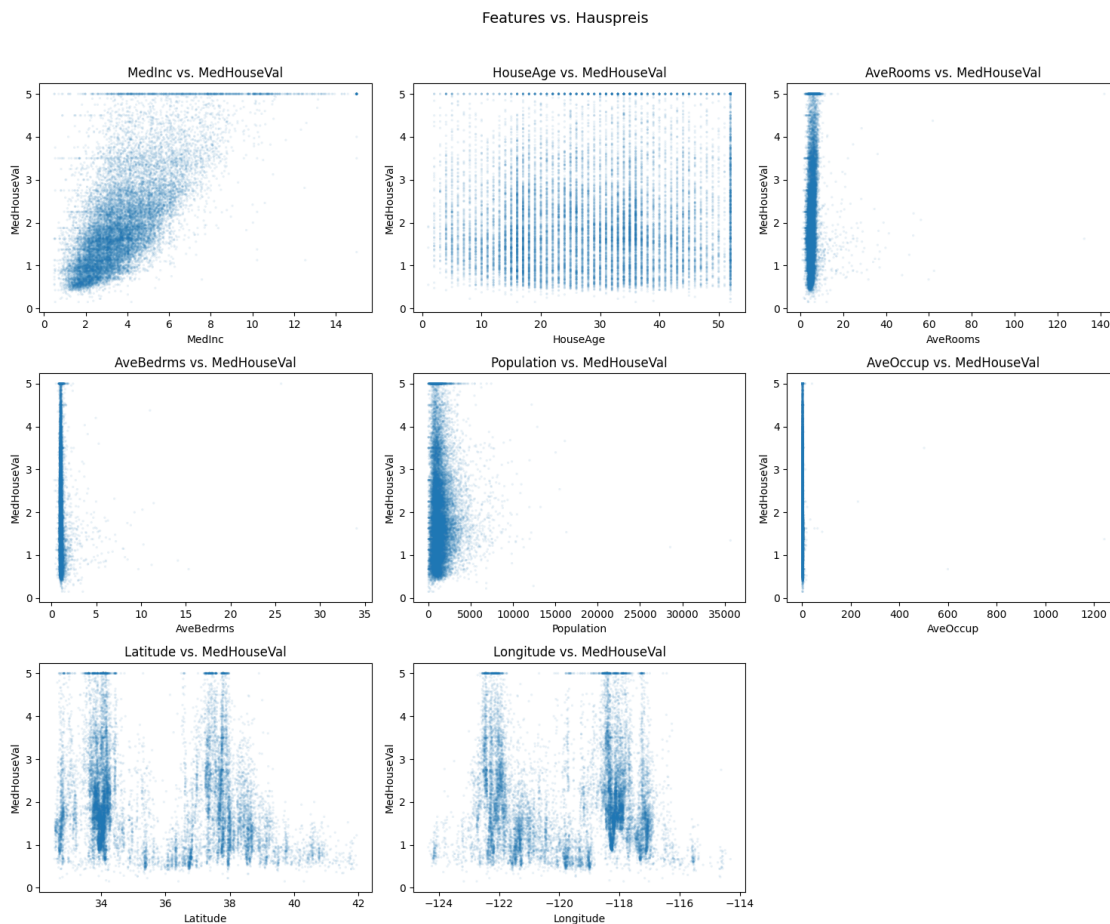


Beobachtungen: - MedHouseVal hat einen Cut-off bei 5.001 (Deckelung) - HouseAge hat einen Cut-off bei 52 (Maximum) - AveRooms, AveBedrms, Population, AveOccup haben starke Outlier

1.3 1.3 Features vs. Hauspreis (Scatterplots)

```
[6]: fig, axes = plot_features_vs_target(df_raw,
    ↪save_name="eda_features_vs_target_raw")
plt.show()
```

Gespeichert: results/eda_features_vs_target_raw.png und
results/eda_features_vs_target_raw.pdf



1.4 1.4 Datenbereinigung

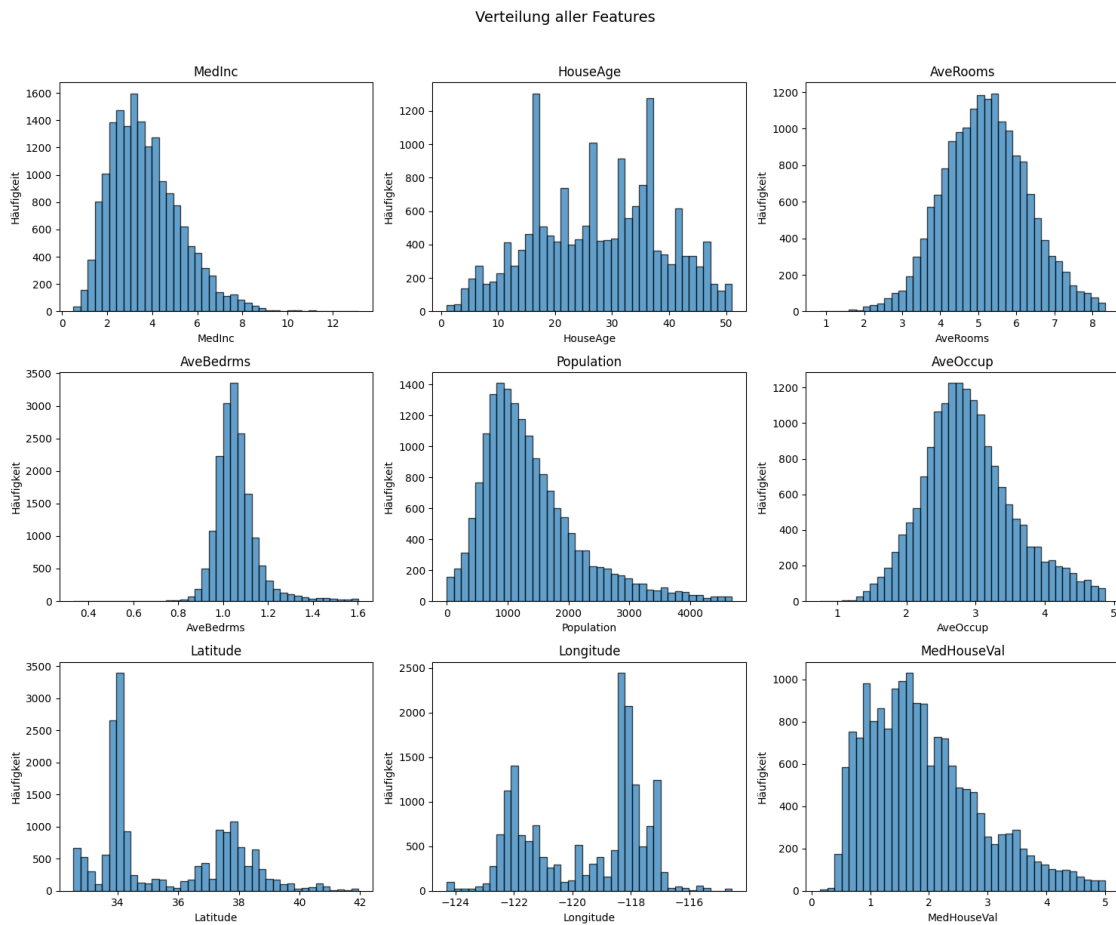
Strategie (nach Vorlage Blatt 10): 1. Cut-off-Werte bei MedHouseVal und HouseAge entfernen 2. Outlier über dem 98%-Quantil bei AveRooms, AveBedrms, Population, AveOccup entfernen

```
[7]: df_clean = clean_data(df_raw)
print(f"Rohdaten:      {df_raw.shape[0]} Zeilen")
print(f"Bereinigt:     {df_clean.shape[0]} Zeilen")
print(f"Entfernt:        {df_raw.shape[0] - df_clean.shape[0]} Zeilen ({(df_raw.
    ↪shape[0] - df_clean.shape[0]) / df_raw.shape[0] * 100:.1f}%)")
```

Rohdaten: 20640 Zeilen
Bereinigt: 17386 Zeilen
Entfernt: 3254 Zeilen (15.8%)

```
[8]: fig, axes = plot_histograms(df_clean, bins=40, save_name="eda_histograms_clean")  
plt.show()
```

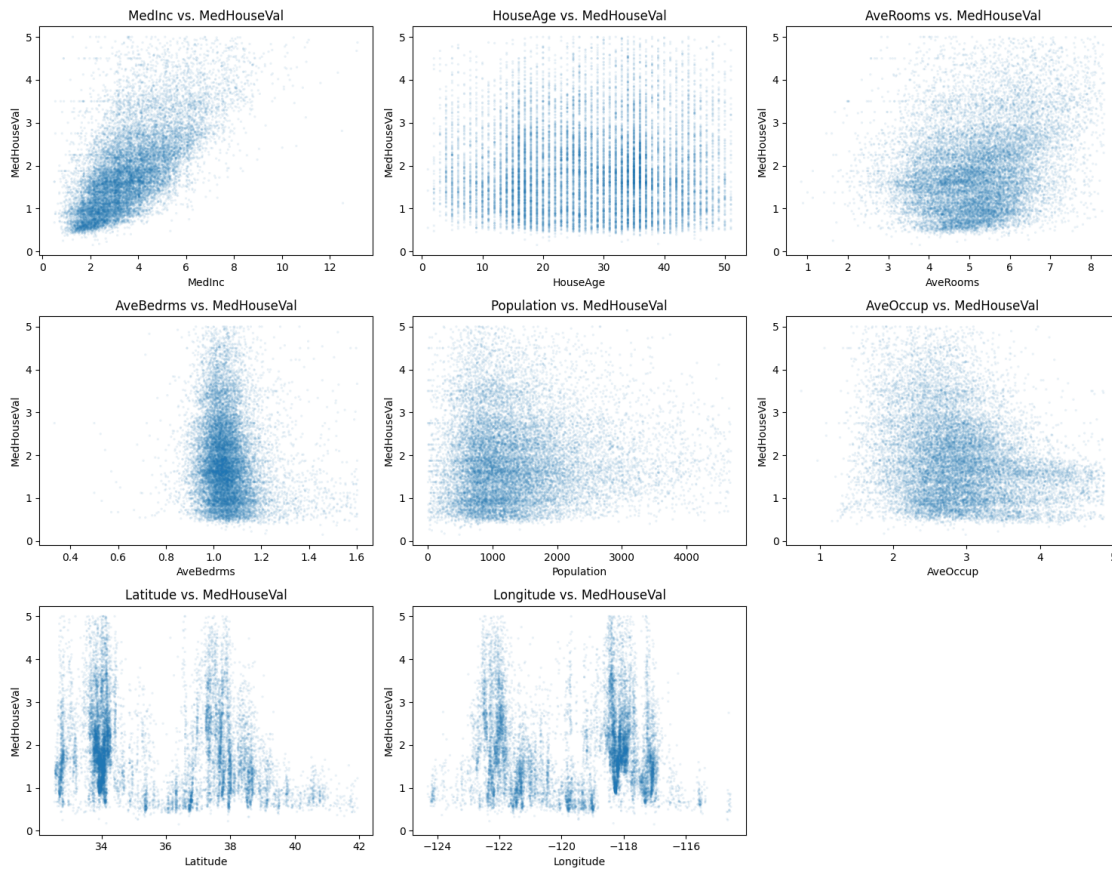
Gespeichert: results/eda_histograms_clean.png und
results/eda_histograms_clean.pdf



```
[9]: fig, axes = plot_features_vs_target(df_clean,   
    ↪ save_name="eda_features_vs_target_clean")  
plt.show()
```

Gespeichert: results/eda_features_vs_target_clean.png und
results/eda_features_vs_target_clean.pdf

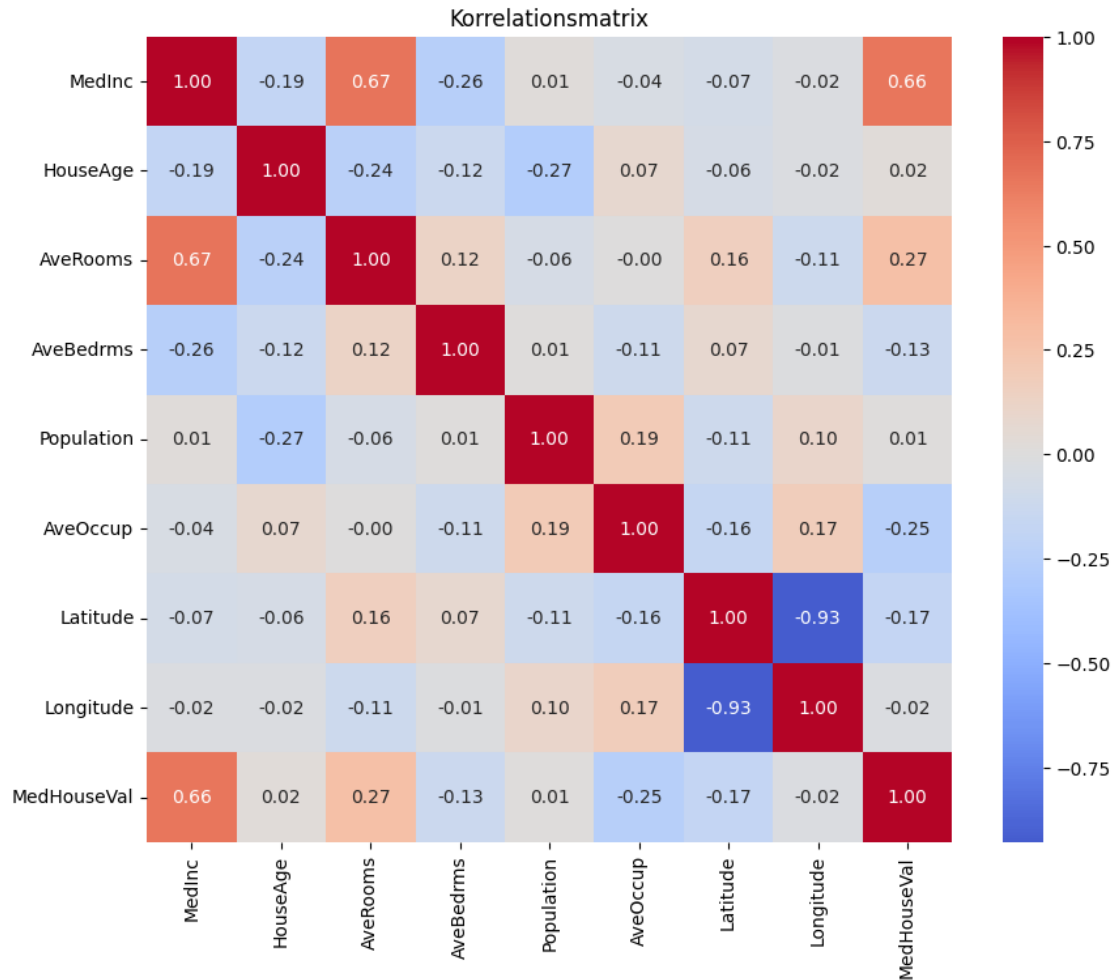
Features vs. Hauspreis



1.5 1.5 Korrelationsanalyse

```
[10]: fig, ax = plot_correlation_heatmap(df_clean,
    ↪ save_name="eda_correlation_heatmap")
plt.show()
```

Gespeichert: results/eda_correlation_heatmap.png und
results/eda_correlation_heatmap.pdf



```
[11]: # Korrelation mit der Zielvariable, sortiert
corr_target = df_clean.corr()["MedHouseVal"].drop("MedHouseVal").
    ↪sort_values(ascending=False)
print("Korrelation mit MedHouseVal:")
print(corr_target.to_string())
```

Korrelation mit MedHouseVal:

```
MedInc      0.659251
AveRooms    0.271879
HouseAge    0.024802
Population  0.012416
Longitude   -0.019278
AveBedrms   -0.127971
Latitude    -0.174653
AveOccup    -0.254479
```

1.6 1.6 Zusammenfassung der EDA

Aspekt	Ergebnis
Datensatzgröße	~20.640 → nach Bereinigung ~17.000+
Stärkste Korrelation	MedInc MedHouseVal
Cut-offs entfernt	MedHouseVal=5.001, HouseAge=52
Outlier-Methode	98%-Quantil auf AveRooms, AveBedrms, Pop, AveOccup
Missing Values	Keine

Nächster Schritt: Baseline mit Linearer Regression erstellen.